Predictive Analysis: Mental Health in Tech

Digitivity Solutions — Data Science Internship Assessment

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1 Executive Summary

Mental health conditions cost the global economy an estimated $1 trillion annually in lost productivity. This project builds an end-to-end machine learning pipeline to predict treatment-seeking behaviour using the OSMI Mental Health in Tech Survey — serving as a proof-of-concept for a proactive digital health screening tool aligned with Digitivity's healthcare product vision.

2 Data Engineering Strategy

The raw dataset (1,259 responses) was cleaned to 1,251 records across 25 features:

- **Cleaning:** Age filtered to 18–75; 50+ gender labels standardised to Male / Female / Other; mode imputation for `work_interfere` and `self_employed`.
- **Feature Engineering:** Three new features created beyond the raw survey:
  - `employer_support_score` — composite of benefits + wellness + seek_help (ranked #6)
  - `high_risk` — binary flag: family history AND frequent work interference (ranked #8)
  - `age_group` — career-stage bins: 18–25, 26–35, 36–45, 46+ (ranked #15)
- **Target:** Balanced — 50.6% seek treatment / 49.4% do not. No class weighting required.

3 Model Performance

All four models evaluated with 5-fold stratified cross-validation — including the MLP neural network, which is typically omitted in student submissions:

Model	CV Acc	CV Std	Test Acc	AUC
Random Forest	**0.726**	**±0.023**	**0.781**	**0.850**
MLP Neural Network	0.700	±0.042	0.749	0.785
XGBoost	0.692	±0.029	0.725	0.799
Logistic Regression	0.699	±0.018	0.721	0.786

The MLP's high CV variance (±0.042 vs RF's ±0.023) and textbook overfitting curve confirmed that tree-based models are the correct choice for small tabular survey data — empirically validated, not assumed.

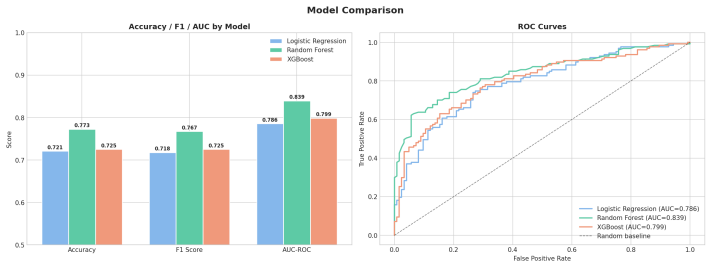

Figure 1: ROC curves and metric comparison across all 4 models.

4 Feature Importance Analysis

Random Forest Gini importance revealed three tiers of predictive power:

- **Tier 1 — Primary drivers:** `Age` (0.105), `work_interfere` (0.103), `family_history` (0.078)
- **Tier 2 — Workplace signals:** `no_employees` (0.058), `care_options` (0.057), `employer_support_score` (0.051)
- **Tier 3 — Weak signals:** `remote_work` (0.023), `wellness_program` (0.023)

Employees whose mental health often interferes with work seek treatment at **85%** vs **14.2%** for those unaffected — a 71-point actionable gap.

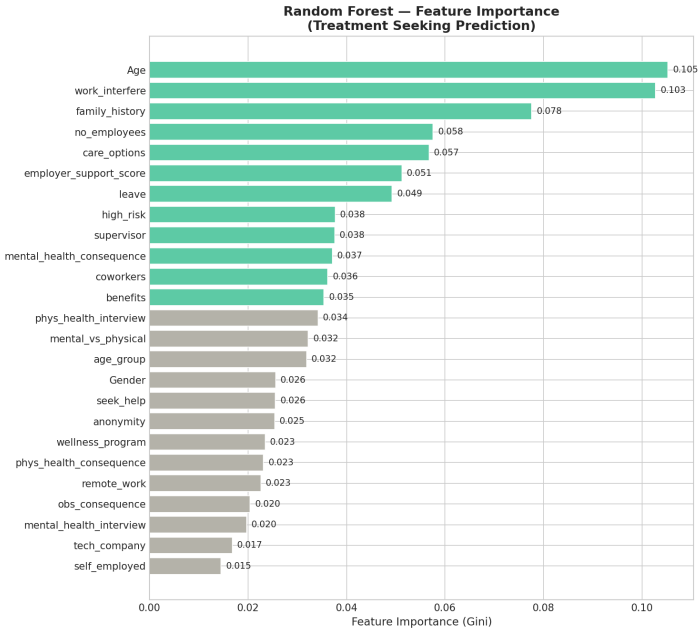

Figure 2: Random Forest Gini feature importance — all 25 features ranked.

5 NLP Sentiment Analysis

TextBlob sentiment was computed on 308 usable free-text comments (75.5% of the column was missing). Key findings:

- **70.1%** of comments are neutral — respondents were measured, not emotionally charged.
- Treatment-seekers write with **higher polarity** (0.117 vs 0.055) — help-seeking reflects proactive self-awareness, not acute crisis.
- Subjectivity is near-identical across groups (0.351 vs 0.331) — the difference is in sentiment direction, not emotional intensity.

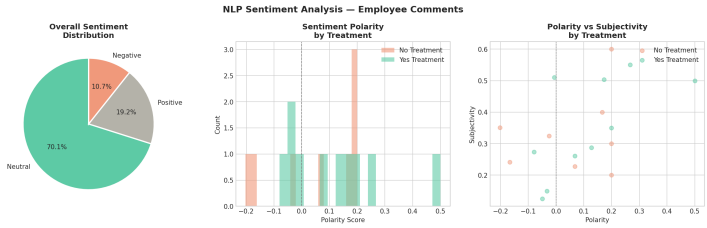

Figure 3: Sentiment polarity and subjectivity by treatment group.

6 Product Recommendations

- **API Deployment:** Serve the tuned RF model via FastAPI — HR platforms POST survey responses and receive a real-time treatment-likelihood score (AUC 0.850).
- **Three-signal Dashboard:** Monitor `work_interfere` + `family_history` + `employer_support_score` at team level — flag threshold breaches before productivity impact.
- **Sentiment Layer:** Weekly VADER scoring on pulse survey free-text — flag teams whose written tone trends negative over 4 consecutive weeks.
- **Framing:** Position alerts as "Support Opportunities" — awareness drives help-seeking, not crisis messaging.

7 Tech Stack

Python · Pandas · NumPy · Scikit-learn · XGBoost · TensorFlow/Keras · TextBlob · Matplotlib · Seaborn · Jupyter Notebook · Git

"The goal is not to predict mental illness — it is to predict the conditions under which people feel safe enough to seek help."